Exploring Deliberative Reasoning for Agentic AR Navigation

Robert Hoeh [*] Jalal Uddin [†] Gabriel Coffee [‡] Jemimah Ogbomoh [§] Peeranat Wattananuruk [¶]
Yu Zhao [‖]

Kennesaw State University, USA

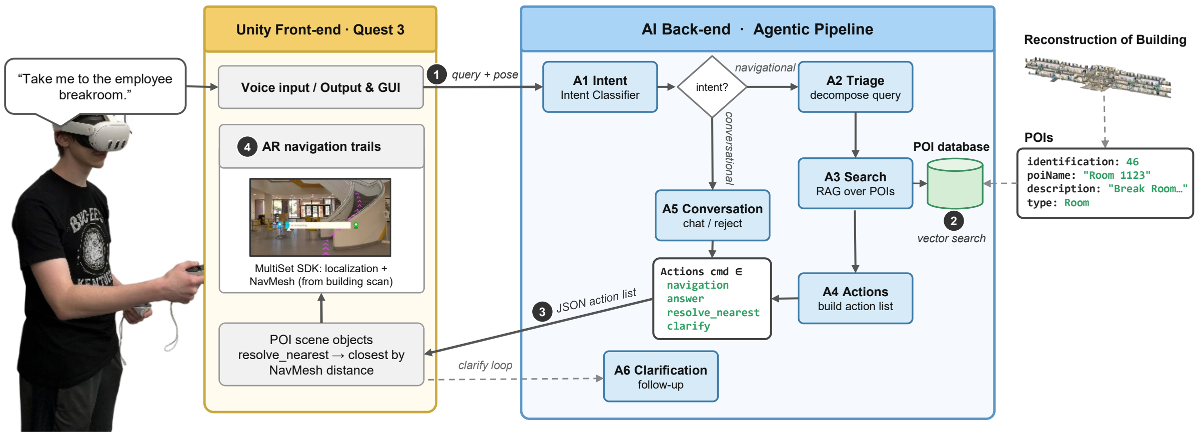


Figure 1: Overview of the proposed agentic AR navigation system. (1) The user issues a voice query through the AR interface, together with the current camera pose. (2) The agentic backend retrieves relevant spatial context from the POI database and coordinates six specialized agents for intent classification, query decomposition, retrieval, action generation, conversational response, and clarification. (3) The backend outputs a structured list of *Agent Actions* that decouples AI reasoning from interface execution. (4) The front-end executes these actions to provide conversational responses and heads-up AR navigation guidance.

Abstract

As conversational AI becomes increasingly integrated into heads-up computing, understanding how intelligent agents should support users during on-the-move interaction is becoming an important research challenge. In this paper, we present an agentic framework for conversational Augmented Reality (AR) navigation that supports intent classification, spatial retrieval, clarification, and action generation through a multi-agent architecture. We conducted a preliminary evaluation using 75 navigation queries with varying ambiguity and complexity to investigate the role of deliberative reasoning. Our results show that enabling thinking mode does not significantly improve response correctness but significantly increases latency for the evaluated tasks, suggesting that lightweight reasoning may be sufficient for routine conversational navigation. This work informs agentic AI design choice and motivates future research on human-AI interaction within AR.

Index Terms: Augmented Reality, Heads-up Computing, Agentic AI, Navigation, Human-AI Interaction.

*e-mail: rhoeh@students.kennesaw.edu
†e-mail: juddin3@students.kennesaw.edu
‡e-mail: gcoffee3@students.kennesaw.edu
§e-mail: jogbomo1@students.kennesaw.edu
¶e-mail: pwattan2@students.kennesaw.edu
‖e-mail: yzhao20@kennesaw.edu

1 Introduction

The heads-up computing paradigm [10] enables Augmented Reality (AR) navigation systems to provide users with navigation assistance while maintaining continuous awareness of the surrounding environment. Research has demonstrated that users with AR navigation systems complete navigation tasks significantly faster, make fewer errors, and report lower anxiety and workload compared to traditional navigation methods [3]. In addition, AR navigation can reduce cognitive load and facilitate the development of cognitive maps [7, 11]. Despite these advances, the interaction paradigms of existing AR navigation systems remain limited. Most systems require users to formulate navigation goals using predefined destinations or structured search interfaces [1, 2]. However, menu-driven interactions can interrupt the flow of navigation, increase attentional demands, and provide limited support for evolving user intent during locomotion, especially for users with physical and cognitive impairment [8].

Recent research in agentic AI offer an opportunity to rethink heads-up navigation as a collaborative interaction between users and intelligent agents. Current advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) provide a promising foundation for these capabilities by enabling natural language interaction while retrieving relevant spatial knowledge to produce contextually grounded responses and make the navigation experience more flexible. Yang et al. [8] combined Building Information Model (BIM) data with a multi-agent RAG framework to support open-ended navigation through an embodied AR agent. Magdy et al. [4] further showed that AR indoor navigation can be combined with conversational information retrieval, and that an LLM-based router can reduce context collision by directing user queries to the most relevant knowledge source. Spatial-RAG [9] demonstrated that combining semantic retrieval with spatial rea-

Table 1: Overview of the agents in the multi-agent system.

Agent	Role
A1.intent_classifier	Classifies the user's query as task-oriented or conversational.
A2.triage_agent	Decomposes complex queries into structured sub-queries and extracts navigation intent.
A3.search_agent	Retrieves relevant POIs and spatial context using vector search in parallel for each sub-query.
A4.actions_agent	Generates structured navigation and interface actions from the retrieved information.
A5.conversation_agent	Generates conversational responses for information-seeking queries that do not require navigation actions.
A6.clarification_agent	Requests additional information when user intent is ambiguous or insufficient for task completion.

soning improves navigation-related question answering. Existing work demonstrates the feasibility of integrating LLMs into AR navigation, but provides limited understanding of how agentic AI should interact with users during heads-up navigation. Questions remain regarding when systems should clarify ambiguous intent, how reasoning should be balanced with interaction responsiveness, and how responsibilities should be distributed across specialized agents while keeping users in the loop.

To address these questions, we present an agentic framework for conversational AR navigation and use it as a platform to explore interaction design questions for heads-up navigation. We conduct a preliminary evaluation of deliberative reasoning across navigation queries with varying levels of ambiguity and complexity by comparing response correctness and end-to-end response time with and without thinking mode enabled. Our results suggest that lightweight reasoning is sufficient for the navigation tasks evaluated, motivating future research on adaptive reasoning, clarification strategies, and human-AI interaction in heads-up navigation.

2 System Design

We designed and developed an AR navigation system (see Figure 1) that translates natural language navigation requests into structured navigation actions while keeping users in the loop throughout the interaction. It supports navigation guidance, conversational point-of-interest (POI) queries, clarification of ambiguous user intent, and multi-step navigation requests through an agentic AI pipeline. The system consists of three primary components: (1) an AR navigation interface with real-time localization that executes these actions in the physical environment, (2) an agentic reasoning pipeline that interprets user queries and generates structured action representations, and (3) an *Agent Actions* data schema for representing navigation and interaction behaviors.

2.1 AR Interface and Localization

The AR navigation and localization function was developed with Unity and MultiSet AI[1] (see Figure 1④). We scanned and reconstructed a complex three-floor campus building and generated a Unity NavMesh to support path planning and navigation. Based on the reconstructed environment, we created a database of 174 Points of Interest (POIs), each containing a unique identifier, name, type, and descriptive metadata (see Figure 1②). The description information stored in these POIs includes room name, capacity, square footage, and other semantic attributes that support spatial reasoning and conversational query resolution.

Users interact with the system through a lightweight heads-up interface that provides access to both a destination list and a voice-based query interface supported by speech-to-text in a Meta Quest 3 headset (see Figure 1①). User queries are sent along with the current camera pose to the backend as input for agent reasoning and navigation planning. Once returned from the backend, the front-end draws an arrow on the floor, leading the user to their destination, given they had a navigational request. These features integrate to keep the user within the environment instead of the device, thus adhering to the main design principle of heads-up computing [10].

2.2 Agentic Pipeline

Conversational heads-up navigation requires the system to interpret user intent, retrieve spatial knowledge, resolve destinations, plan routes, and clarify ambiguous requests. To support these distinct interaction responsibilities, we adopt an agentic architecture that decomposes these responsibilities across specialized agents. Our implementation was inspired by the multi-agent framework proposed by Yang et al. [8] and is developed using Google's ADK[2]. The architecture is further motivated by prior work demonstrating the effectiveness of agentic workflows for complex reasoning [6] and RAG for grounding responses in spatial knowledge [5, 9]. The system consists of six specialized agents (Table 1, Figure 1), each responsible for a distinct interaction task: **A1**.intent classification, **A2**.query decomposition, **A3**.POI retrieval, **A4**.action generation, **A5**.conversational response generation, and **A6**.ambiguity clarification. This modular design enables extensible and conversational AR navigation while allowing each agent to specialize in a specific reasoning task.

Agent Actions. To better support human-in-the-loop interaction, we introduce an *Agent Actions* schema that translates high-level agent decisions into structured actions executed by the AR interface (see Figure 1③). The agentic backend outputs a sequence of JSON-based actions that informs the front-end *what* data to resolve, *how* to resolve it, and in *what order*. From there, the front-end resolves each action according to its defined priority and interaction dependencies (e.g., clarification before navigation). The current system supports four action types:

1. `navigation`: Used when a single POI has resolved. Displays AR navigation cues.
2. `resolve_nearest`: Used when there are several favorable POIs, and path distance needs to be taken into account. This action resolves in the background, and uses the Unity scene's NavMesh to find the nearest POI to either the user or the POI before this action.
3. `answer`: Presents conversational responses to information-seeking queries without initiating navigation.
4. `clarify`: Used when the POI the user queries is either ambiguous or could not be found within the database. When resolving, asks the user if they wish to be guided to a list of suggested POIs based on the vector search output and the user's query. This action is resolved using the `clarification_agent`.

The *Agent Actions* schema decouples agent reasoning from interface execution, enabling extensible interaction behaviors while maintaining a consistent mechanism for coordinating AI reasoning and AR navigation.

3 Exploratory Evaluation

To conduct an initial evaluation of the proposed system, we investigated *how deliberative reasoning affects the performance of agentic*

[1]https://www.multiset.ai/

[2]https://docs.cloud.google.com/gemini-enterprise-agent-platform/build/adk

Table 2: Navigation query taxonomy used in the evaluation benchmark.

Ambiguity	Complexity	Validity	Interaction Challenge	Example
Explicit	Single-step	Valid	Goal understanding	“Take me to room 1050.”
Implicit	Single-step	Valid	Intent inference	“Take me to the employee break room.”
Explicit	Multi-step	Valid	Task planning	“Take me to the bathroom, then the nearest conference room.”
Implicit	Multi-step	Valid	Intent inference + task planning	“Find somewhere quiet to study, then guide me to the nearest exit.”
–	–	Invalid	Error detection and recovery	“Take me to the swimming pool on the fourth floor.”

AR navigation. We developed a benchmark consisting of 75 navigation queries spanning five representative query types, each designed to evaluate a distinct interaction challenge encountered in conversational AR navigation (Table 2). The benchmark systematically varies ambiguity, task complexity, and destination validity to assess the system’s ability to support diverse navigation scenarios. It includes 15 queries in each of the five categories: explicit single-step, implicit single-step, explicit multi-step, implicit multi-step, and erroneous.

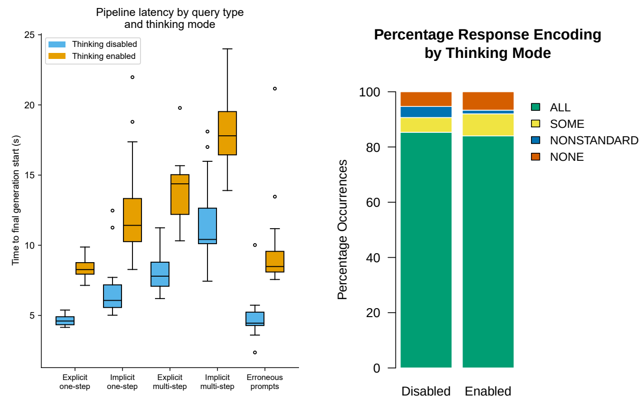
The query set was initially generated using the LLM Claude Sonnet 4.6 and iteratively refined by the authors. The LLM was provided with the system prompt and a JSON representation of the POI database during query generation, reducing the likelihood of generating infeasible or hallucinated navigation requests, except for erroneous queries, where this was wanted. The final test set was manually reviewed to ensure that all queries were appropriate for the target environment and evaluation objectives.

We evaluated the proposed multi-agent system using Gemini-2.5 Flash[3] hosted on Google’s cloud platform under two configurations: with and without thinking mode enabled. For each query, we executed the complete navigation pipeline using the ADK evaluation framework and collected system outputs, response latency, token usage, and agent execution logs. The resulting data were exported and analyzed in R to compare system behavior across the two reasoning configurations.

Response Time. We measured end-to-end system response time as the elapsed time between the last and first timestamp found within each query. The distribution violated the assumption of normality according to the Shapiro–Wilk test (thinking disabled: $W = 0.859$, $p < .001$; thinking enabled: $W = 0.906$, $p < .001$). Therefore, we conducted a Mann–Whitney U test, with an alpha cutoff of 0.05. The results showed a statistically significant increase in end-to-end system response time between the thinking-disabled and thinking-enabled conditions (Mann–Whitney U test: $U = 832$, $p < 0.001$). The mean system response time was $M = 7.62$ s ($SD = 3.29$ s) with thinking disabled and $M = 12.60$ s ($SD = 4.41$ s) with thinking enabled. As illustrated in Figure 2a, the response latency was consistently higher across all query categories for thinking enabled, suggesting that enabling thinking mode introduces noticeable latency differences for the evaluated navigation tasks.

Correctness. To evaluate response correctness, each system response was paired with its corresponding POIs and manually coded into one of four categories: **ALL**, indicating that all sub-queries within the query were successfully answered with *reasonable output*; **SOME**, indicating that only part of the sub-queries received a *reasonable output*; **NONE**, indicating that no sub-query received a *reasonable output*; and **NONSTANDARD**, indicating that the response deviated from expected conventions while still containing partially useful information. A *reasonable output* refers to a response that appropriately addresses the user’s request given the available knowledge base, even when the correct behavior was to indicate that the requested destination could not be found. Figure 2b summarizes the correctness results. The thinking-disabled configuration correctly resolved 64 of 75 queries (85.3%), while the thinking-enabled configuration correctly resolved 63 of 75 queries (84.0%). A two-sample proportion test ($\alpha = .05$) found no statistically significant difference between the two conditions ($\chi^2 = 0$, $p = 1$, with continuity correction). These results suggest that, for the navigation tasks evaluated in this study, enabling thinking mode did not significantly improve response correctness.

[3]https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash



(a) Pipeline latency by query type

(b) Response encoding by thinking mode

Figure 2: Pipeline latency and response correctness across thinking conditions.

4 Implications and Future Work

Our preliminary evaluation shows that enabling thinking mode did not significantly improve response correctness, while significantly increasing response time. This suggests that lightweight reasoning may be sufficient for the navigation tasks evaluated in this work while potentially reducing computational cost [12]. Future work will investigate this trade-off by measuring per-agent latency and token usage and by evaluating more complex navigation scenarios, such as ambiguity resolution and multi-turn interactions.

Future user studies will investigate how AI actions should be coordinated to support natural and efficient heads-up interactions, including when systems should proactively request clarification versus autonomously resolving user intent. We also plan to study how users interact with agentic navigation over extended conversations, including how conversational context and evolving navigation goals should be managed, and how exposing different levels of agent reasoning influences user understanding, trust, cognitive workload, and the overall navigation experience. In addition, future work will investigate interaction modalities beyond voice, such as hand gestures and gaze, to support more efficient and natural heads-up computing.

Finally, we plan to extend the underlying spatial representation beyond a vector-based POI database by incorporating scene graph representations of hallway-centric indoor environments together with mechanisms for dynamic information updates. Richer spatial representations can provide stronger grounding for LLM-based reasoning and enable the system to adapt more effectively to changes in real-world environments.

ACKNOWLEDGMENTS

This work was supported in part by grants from the KSU First-Year Scholars program and the Summer Undergraduate Research Program.

REFERENCES

[1] M. Gehring and P. Mosler. Indoor navigation with augmented reality and bim: A marker-based approach for locating logistics areas on construction sites. 09 2022.

[2] B.-C. Huang, J. Hsu, E. T.-H. Chu, and H.-M. Wu. ARBIN: Augmented reality based indoor navigation system. Sensors (Basel), 20(20), Oct. 2020.

[3] K. Liu, M. L. Aebersold, M. Lindquist, and H. Gao. Augmented reality indoor wayfinding in hospital environments an empirical study on navigation efficiency, user experience, and cognitive load, 2025.

[4] A. Magdy, A. Mohamed, A. A. Boghdady, B. Achraf, H. Mohamed, Z. Abd-El-Hamid, S. Ali, and I. Hegazy. A unified framework for ar indoor navigation and conversational information retrieval in smart campus environments. In 2025 Twelfth International Conference on Intelligent Computing and Information Systems (ICICIS), pp. 757–763, 2025. doi: 10.1109/ICICIS66182.2025.11313163

[5] S. C. Marcelyn, Y. Gao, Y. Zhang, and X. Gao. Pathgpt: Reframing path recommendation as a natural language generation task with retrieval-augmented language models, 2025.

[6] C. Nandkumar and L. Peternel. Enhancing supermarket robot interaction: an equitable multi-level llm conversational interface for handling diverse customer intents. Frontiers in Robotics and AI, 12, Apr. 2025. doi: 10.3389/frobt.2025.1576348

[7] Z. Qiu, A. Mostafavi, and S. Kalantari. Use of augmented reality in human wayfinding: a systematic review. Virtual Reality, 29(4):154, Sept. 2025.

[8] H.-K. Yang, T.-C. Hsiao, R. Oka, R. Nishino, S. Tofukuji, and N. Kobori. An embodied ar navigation agent: Integrating bim with retrieval-augmented generation for language guidance. In 2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), pp. 1461–1471, 2025. doi: 10.1109/ISMAR67309.2025.00150

[9] D. Yu, R. Bao, R. Ning, J. Peng, G. Mai, and L. Zhao. Spatial-RAG: Spatial retrieval augmented generation for real-world geospatial reasoning questions. In M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, eds., Findings of the Association for Computational Linguistics: ACL 2026, pp. 11094–11112. Association for Computational Linguistics, San Diego, California, United States, July 2026. doi: 10.18653/v1/2026.findings-acl.539

[10] S. Zhao, F. Tan, and K. Fennedy. Heads-up computing moving beyond the device-centered paradigm. Commun. ACM, 66(9):56–63, Aug. 2023. doi: 10.1145/3571722

[11] Y. Zhao, H. Gagnon, J. Stefanucci, S. Creem-Regehr, and B. Bodenheimer. A systematic review of the use of augmented reality in pedestrian navigation. ACM Computing Surveys, 58(5):1–32, 2025.

[12] S. Zhou, R. Ling, J. Chen, X. Wang, T. Fan, and H. Wang. When more thinking hurts: Overthinking in llm test-time compute scaling, 2026.